WORKED SOLUTION

STEP I 2011, Question 7

Pure • Separable Differential Equations

STEP I 2011, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s1-q7

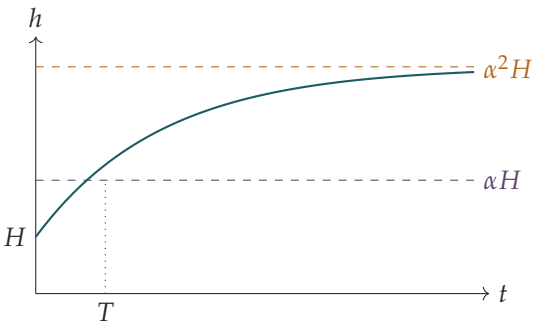
The key idea

Volume equals (constant area) $\times$ height, so the rate of change of height is just (inflow $-$ outflow) divided by the area. The phrase "the height remains constant" is the clue that pins down the inflow: at that height, inflow and outflow balance exactly. After that, each part is a separable differential equation, and in part (ii) the substitution $u = \sqrt{h}$ turns the integral into something routine.

Solution

Let the constant cross-sectional area be A and the constant inflow rate be q (volume per unit time). The volume of water is Ah, so

$$A\frac{\mathrm{d}h}{\mathrm{d}t} = q - (\text{outflow rate}).$$



The height rises from H towards the level $\alpha^2 H$ at which inflow and outflow balance, getting ever closer but never reaching it; the question is how long it takes to get to αH, which lies between them because $1 < \alpha < \alpha^2$. (The sketch shows part (i).)

Part (i)

The outflow rate is λh for some constant $\lambda > 0$, so

$$A\frac{\mathrm{d}h}{\mathrm{d}t} = q - \lambda h.$$

When $h = \alpha^2 H$ the height stays constant, so $\mathrm{d}h/\mathrm{d}t = 0$ there, giving $q = \lambda\alpha^2 H$. Substituting,

$$\frac{\mathrm{d}h}{\mathrm{d}t} = \frac{\lambda}{A}\left(\alpha^2 H - h\right) = k\left(\alpha^2 H - h\right), \qquad k = \frac{\lambda}{A} > 0,$$

as required.

Since $H \le h < \alpha^2 H$ throughout the rise, $\alpha^2 H - h > 0$ and we may separate the variables:

$$\begin{aligned} kT &= \int_H^{\alpha H} \frac{\mathrm{d}h}{\alpha^2 H - h} = \Big[-\ln\left(\alpha^2 H - h\right) \Big]_H^{\alpha H} = \ln\frac{\alpha^2 H - H}{\alpha^2 H - \alpha H} \\ &= \ln\frac{(\alpha-1)(\alpha+1)}{\alpha(\alpha-1)} = \ln\frac{\alpha+1}{\alpha} = \ln\left(1 + \frac{1}{\alpha}\right), \end{aligned}$$

cancelling $\alpha - 1 \neq 0$ because $\alpha > 1$. This is the required result.

For large α, $x = 1/\alpha$ is small, so

$$kT \approx \frac{1}{\alpha} - \frac{1}{2\alpha^2} \approx \frac{1}{\alpha},$$

since the second term is negligible compared with the first when α is large.

Part (ii)

Now the outflow rate is $\mu\sqrt{h}$ for some constant $\mu > 0$, so $A\,\mathrm{d}h/\mathrm{d}t = q - \mu\sqrt{h}$. The height is constant at $h = \alpha^2 H$, so $q = \mu\sqrt{\alpha^2 H} = \mu\alpha\sqrt{H}$ (using $\alpha > 0$). Hence

$$\frac{\mathrm{d}h}{\mathrm{d}t} = c\left(\alpha\sqrt{H} - \sqrt{h}\right), \qquad c = \frac{\mu}{A} > 0.$$

Again the bracket is positive while $h < \alpha^2 H$, so

$$cT' = \int_H^{\alpha H} \frac{\mathrm{d}h}{\alpha\sqrt{H} - \sqrt{h}}.$$

? **Ask yourself** How do we deal with the $\sqrt{h}$ in the denominator?

Substitute $u = \sqrt{h}$, so $h = u^2$ and $\mathrm{d}h = 2u\,\mathrm{d}u$; the limits become $u = \sqrt{H}$ and $u = \sqrt{\alpha H}$. Writing $a = \alpha\sqrt{H}$ for short, the integrand is a rational function, and dividing out,

$$\frac{2u}{a-u} = -2 + \frac{2a}{a-u}.$$

Therefore

$$\begin{aligned} cT' &= \int_{\sqrt{H}}^{\sqrt{\alpha H}} \left(-2 + \frac{2a}{a-u}\right) \mathrm{d}u = \Big[-2u - 2a\ln(a-u)\Big]_{\sqrt{H}}^{\sqrt{\alpha H}} \\ &= -2\left(\sqrt{\alpha H} - \sqrt{H}\right) + 2a\sqrt{H}\,\ln\frac{a\sqrt{H} - \sqrt{H}}{a\sqrt{H} - \sqrt{a}\sqrt{H}} \\ &= 2\sqrt{H}\left(1 - \sqrt{\alpha} + \alpha\ln\frac{\alpha-1}{\alpha-\sqrt{\alpha}}\right). \end{aligned}$$

The logarithms are of positive quantities because $u \le \sqrt{\alpha H} < \alpha\sqrt{H} = a$ (as $\sqrt{\alpha} < \alpha$ for $\alpha > 1$). Finally, factorising $\alpha - 1 = (\sqrt{\alpha}-1)(\sqrt{\alpha}+1)$ and $\alpha - \sqrt{\alpha} = \sqrt{\alpha}(\sqrt{\alpha}-1)$,

$$\frac{\alpha-1}{\alpha-\sqrt{\alpha}} = \frac{\sqrt{\alpha}+1}{\sqrt{\alpha}} = 1 + \frac{1}{\sqrt{\alpha}},$$

so

$$cT' = 2\sqrt{H}\left(1 - \sqrt{\alpha} + \alpha\ln\left(1 + \frac{1}{\sqrt{\alpha}}\right)\right),$$

as required.

For large α, $1/\sqrt{\alpha}$ is small, so

$$\alpha\ln\left(1 + \frac{1}{\sqrt{\alpha}}\right) \approx \alpha\left(\frac{1}{\sqrt{\alpha}} - \frac{1}{2\alpha}\right) = \sqrt{\alpha} - \frac{1}{2}.$$

The large terms $\pm\sqrt{\alpha}$ cancel, leaving

$$cT' \approx 2\sqrt{H}\left(1 - \sqrt{\alpha} + \sqrt{\alpha} - \tfrac{1}{2}\right) = \sqrt{H}.$$

Notice that we needed the x^2 term of the logarithm here: with only $\ln(1+x) \approx x$, the bracket would be 1 and we would get $2\sqrt{H}$, which is wrong. That is why the question gives the two-term approximation.

Checks

- As $\alpha \to 1^+$, (i) gives $kT \to \ln 2$, not 0. That is right: the gap to equilibrium starts at $(\alpha^2-1)H \approx 2(\alpha-1)H$ and must shrink to $(\alpha^2-\alpha)H \approx (\alpha-1)H$, so the gap halves, and an exponential decay at rate k halves in time $(\ln 2)/k$. In (ii), $cT' \to 2\sqrt{H}\ln 2$, which matches the same halving with the effective rate $c/(2\sqrt{H})$ obtained by linearising $\sqrt{h}$ near $h = H$.
- With $\alpha = 4$ and $H = 1$: $kT = \ln 1.25 \approx 0.2231$ and $cT' = 2(-1 + 4\ln 1.5) \approx 1.2437$. Stepping both differential equations numerically (Runge–Kutta) for several random H, α, k, c reproduces both formulae to 10^{-9}.
- For $\alpha = 10^4$, $H = 2$: $\alpha\ln(1 + 1/\alpha) \approx 0.99995$ and $cT'/\sqrt{H} \approx 1.0066$, consistent with the limits (the next term gives $cT' \approx \sqrt{H}\,(1 + \frac{2}{3\sqrt{\alpha}})$).

Common slips

- Forgetting to use the equilibrium condition to eliminate the inflow rate, and so being left with too many constants.
- Using only $\ln(1+x) \approx x$ in part (ii), which loses the crucial $-\frac{1}{2}$.
- Writing $\sqrt{\alpha^2 H}$ as $\alpha^2\sqrt{H}$.

Beyond STEP

Both models are first-order autonomous differential equations $\dot{h} = f(h)$ with a stable equilibrium at $h = \alpha^2 H$; the linear case is Newton's law of cooling in disguise, while the $\sqrt{h}$ case is Torricelli's law for flow through a hole. Studying equilibria and their stability without solving explicitly is the start of the qualitative theory of ODEs. Boyce and DiPrima, *Elementary Differential Equations*, covers autonomous equations and tank-mixing models in its early chapters.

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